The Red Queen 120-cell

Plan: the polar exchange of the White Queen 600-cell



DICTO · idea 27–28 October 2026: "a polar swap of the White Queen", "can we build the polar exchange 120 cell" · study 2026-10-27-white-queen-polar

> **Goal.** Build the 120-cell, the polar of the 600-cell, exactly, and show the exchange between the two: every vertex of one becomes a cell of the other. The White Queen has 600 tetrahedra round 120 corners; her polar, the Red Queen, named after the other queen of *Through the Looking-Glass*, has 120 dodecahedra round 600 corners. Result wanted: the same proved atlas as #63 for the Red Queen, the exact dictionary between the two, and a Polar button in the White Queen world.

Already proved by our own data

The corners of the Red Queen are exactly the centres of the White Queen's 600 cells. So the Cleo shells of #63 are already a result about the Red Queen:

White Queen 600-cell (#63)	Red Queen 120-cell
600 cells in 15 shells round one vertex: 20, 20, 30, 60, 60, 60, 20, 60, 20, 60, 60, 60, 30, 20, 20	600 corners in the same 15 layers round one dodecahedral cell
Core: 20 Cleo-Icos filling a regular icosahedron	Core: the 20 corners of the central dodecahedron
120 vertices in 9 layers round a vertex: 1, 12, 20, 12, 30, 12, 20, 12, 1	120 dodecahedra in the same 9 layers round one dodecahedron
The list reads the same backwards (central symmetry)	The same, for the same reason

So the polar exchange of the Cleo core is the dodecahedron itself: twenty Cleo-Icos round a point on one side, twenty corners round a cell on the other. This is the icosahedron–dodecahedron duality of the Platonic solids, one dimension up.

Stages

Stage	Work	Where	Cost
0	Prior art: Coxeter, *Regular Polytopes*; Alicia Boole Stott's 120-cell sections (about 1900); Schlegel diagrams and physical 120-cell models; every source checked listed.	library + web	small
1	Exact source model of the 120-cell from the #63 source model: 600 corners, 1200 edges, 720 pentagons, 120 cells, every cell a regular dodecahedron; checks as for #63.	dicto-big-brother	small, no tokens while it runs
2	Red Queen shells: what each of the 9 dodecahedron layers looks like in perspective, and whether each layer is one kind of piece, as the Cleo shells are.	dicto-big-brother	small
3	Red Queen atlas: slices and four shadows in the four main directions, exact, as #63; the halves of each shadow.	dicto-big-brother	medium
4	The dictionary: each White Queen slice, shadow and shell beside its polar, in one table and one figure.	study	small
5	App: a Polar button in the White Queen world swaps 600-cell and 120-cell in the Shadow, Slice and Shells views; the gold Cleo core turns into the gold central dodecahedron.	Kaleidohedra	medium
6	Paper #64 (placeholder): The Red Queen 120-cell, led by its main theorem, the exact polar dictionary.	paper	medium

Stages 0–2: results (28 October 2026)

> **Verdict.** The Red Queen is built exactly, and the polar exchange holds point for point. 13 of 13 exact checks pass (scripts/red_queen.py, data/red_queen.json).

Check	Result
600 corners (the White Queen's 600 cell centres), 1200 equal edges, 4 at every corner	exact
720 regular pentagons (one round each White Queen edge), each in exactly 2 cells	exact
120 cells (one round each White Queen vertex), each a regular dodecahedron	exact
Dodecahedra round one cell: 9 layers 1, 12, 20, 12, 30, 12, 20, 12, 1	exact (known, see below)
Corners round one cell: 15 layers 20, 20, 30, 60, 60, 60, 20, 60, 20, 60, 60, 60, 30, 20, 20, the Cleo shells	exact
In perspective from just beyond the opposite cell, every dodecahedron layer is one kind of piece	exact; it follows from the icosahedral symmetry round the axis, which turns each layer onto itself
The core: the central dodecahedron, its 20 corners the centres of the 20 Cleo-Icos	exact
Its 12 pentagons are the polar of the 12 corners of the Cleo core's icosahedron	exact

The pentagonal pyramids. Cut from its centre, the Red Queen's core falls into the 12 pentagonal pyramids of the regular dodecahedron, Aristotle's dodecahedron pyramid of #44, as the White Queen's core falls into the 20 Cleo-Icos. The dictionary of the two cores: Cleo-Icos ↔ corner of the dodecahedron (20 ↔ 20); corner of the icosahedron ↔ pentagonal pyramid (12 ↔ 12). In the 120-cell the core is one whole cell; the 12 pyramids are its cut from the centre, not cells of their own.

Sources checked (stage 0, 28 October 2026). Reference library (ask-library on "120-cell layers round one cell; Boole Stott sections; Schlegel diagram; 600-cell / 120-cell duality"): H. S. M. Coxeter, *Regular Polytopes*, 3rd ed. (1973); M. Koca, N. Koca, M. Al-Ajmi, "Branching of the W(H4) polytopes" (2011); M. Koca, M. Al-Ajmi, N. Koca, grand antiprism by quaternions (2009); Kaida Shi (2009), overall shapes of the 120-cell and 600-cell; A. Garrison, R. Scott, "Small covers of the dodecahedron and the 120-cell" (2001). Web search "120-cell layers of cells around one dodecahedron 1, 12, 20, 12, 30": Wikipedia "120-cell" (cites J. Stillwell, "The story of the 120-cell", Notices AMS 2001); S. Schleimer and H. Segerman, "Puzzling the 120-cell", arXiv 1310.3549, and their sculpture talks, which give the layers 1, 12, 20, 12, 30, … round one dodecahedron; the Zometool 120-cell model page (Ruhr-Universität Bochum). The 9 layers are known; the reading of the Cleo shells as the Red Queen's corner layers, and the core dictionary with Aristotle's pyramids, were not found in these sources.

Added to the Red Queen paper (DICTO 28 October 2026)

Main theorem: Aristotle's five pyramids are the cores of the regular four-dimensional solids. Tetrahedron, octahedron and icosahedron pyramids: the shadows of the cells round one corner of the 5-cell, the 16-cell and the 600-cell. Cube, octahedron and dodecahedron pyramids: the shadows of the sides of the 4D pyramid from the centre over one cell of the tesseract, the 24-cell and the 120-cell. Polar pairs: 16-cell (8) ↔ tesseract (6); 600-cell (20) ↔ 120-cell (12); the 5-cell its own polar. To be checked exactly for all six.

The star core. No convex regular 4D solid has dodecahedra round a corner ({3,5,3} is the infinite icosahedral honeycomb of hyperbolic space). The small stellated 120-cell {5/2,5,3} (Hess 1883; Coxeter) does: 120 small stellated dodecahedra on the 600-cell's 120 corners, 1200 edges, 720 pentagrams, a regular dodecahedron round every corner. Each of the 12 cells at a corner meets it with one spike, a pentagonal pyramid; to be checked exactly whether these 12 spikes form the dodecahedron cut into Aristotle's 12 pyramids. Also: the 4D pyramid over a dodecahedron has its 12 pentagonal pyramids round the tip, but never with all edges equal (circumradius 1.401 edges), unlike the icosahedron (0.951), which is why the 600-cell closes and nothing equal-edged closes round a dodecahedron.

Order

Stages 0–2 now: they run on dicto-big-brother and cost few tokens, so they fit the frugal week. Stages 3–6 after the frugal week, after the build pieces (Icosaxes, Lotus Jewel Axes, Unborn Axes), which stay first.

Name: The Red Queen 120-cell (DICTO 28 October 2026). Carroll references stay in the White Queen and Red Queen papers only.